

Stuart Small

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Seasoned security leader with over 15 years of experience building and scaling engineering and security programs from the ground up, particularly in early-stage startups. With a career starting in software develop ranging from device drivers all the way up to web frontends. I have a deep knowledge of secure code practices, low level networking, architecture, and efficient software development lifecycle. An expert with the Rust programming language starting soon after the first stable release and including open source work as a member of the Rust Secure Code Working Group. Adept at hands-on technical implementation with the long term eye on designing scalable processes that drive organizational growth and successful exits.

Professional Experience

Ditto May 2021 - Present
Director of Security Operations

- Started as an engineer with focus on our kubernetes based cloud portion of our offline-first, peer-to-peer, CRDT database. Implemented the authentication system, wrote the current HTTP interface and many other features
- Made major improvements to the consensus protocol for our cloud data store services to increase availability during a degraded state.
- Developed series of property based tests for automated correctness testing across large sections of the code base
- Promoted to manager leading the security and QA teams. Eventually promoted to head to security. Ultimately responsible for security architecture, compliance, operational security and customer relations
- Lead push for CMMC compliance while also maintaining and improving existing SOC2 compliance
- Spearheaded modernization and automation of security operations across commercial and defense divisions. This was done through both traditional SOAR methods and AI driven automations
- Worked in a direct customer facing role supporting Sales and Customer Support teams in addressing customer security concerns and explaining the product
- Critical part role in the ATO process for cleaning applications for use in the defense space

Randori May 2020 - May 2021
Principal Engineering Lead

- Built up and lead the team building out the Continuous Automated Red Team (CART) product that leveraged Randori's existing ASM capabilities
- Took over the initial MVP and stabilized it. Added CI, monitoring, wrote test plans and polished the deployment process
- Built new automated phishing functionality and client software reconnaissance feature
- Worked directly with the attack team to identify needs and implement them to maximize their efficiency
- Developed in Python and Go on top of an in house Docker orchestration framework

ThreatX, Inc. June 2016 - May 2020
Director of Engineering

- Joined company during seed funding phase as an early software engineering hire. Took an MVP product and helped transform it into a production worthy, scalable service
- Served in various technical roles inside the company. The primary focus was software engineering but also played a major role in operations, was part of external penetration testing contracts, ran simulated phishing campaigns, and handled customer support
- Actively involved in the development of every part of the product. Technologies regularly used include Rust, MongoDB, Nginx, Consul, RabbitMQ, Typescript, AngularJS, Python3, and Flask. The vast majority of the work was done in Rust.
- Stepped into a leadership position after the series A funding. Led the hiring process in engineering, maintained the roadmap and guided the team towards making the best product possible. Stepped in as scrum master, product owner, and product manager as needed

- Engineer working on a new SIP based smart phone. Worked on almost every portion of the system from the WiFi device driver, Android middleware, bug fixes in upstream AOSP, to writing some user applications. Worked in C, C++, Java and Scala.
- Served as the security expert for the project. Maintained SELinux policies, tracked upstream projects and backported security fixes. Regularly audited parts of the system and wrote PoC exploit code. Wrote blog posts and technical bulletins to explain security issues to customers
- Served as scrum master for the platform team. In charge of keeping the team organized in the context of the sprint, tracking and interpreting metrics, and organizing and running sprint ceremonies
- Rearchitected provisioning system to allow for a fast, secure, zero touch method that works across our product line. A proof of concept was written using Scala on the Play Framework

- Co-founding member of small bootstrapped startup.
- Primarily focused on the mobile strategy. Played major role in design and architecture of the system. Other primary responsibilities included testing, fund raising, and operations

- Member of the OnPlus team. Worked towards making a cloud based network management tool for managing both Cisco and 3rd party network equipment.
- Development time split between an embedded Linux based network discovery device and development of server side services based on FreeBSD, PostgreSQL and PHP.
- Build device emulator to cheaply simulate fleets of devices for correctness and load testing.

- Embedded systems engineering working on a new uClinux based inventory scanner
- Assisted in porting an existing DR-DOS C application, wrote Linux device drivers, validated hardware, automated systems for updating device firmware.

- Web Developer working across two projects. One was PHP, MySQL, Linux and the other being C#, ASP.NET and SQL Server

Publications and Presentations

- Intro to Android Reverse Engineering
- Dependency Injection in Rust
- Client Fingerprinting with TLS Session Resumption with patched rustls
- Building Malware in Rust

Education and Certifications

Bachelor of Science in Computer Science - University of Texas at Dallas

CISSP